Machine Learning Assignment 2

FIFA Players Performance Analysis

In this assignment, you will use a real-world FIFA players dataset to solve three machine learning problems using the same data and preprocessing pipeline:

- Regression **— predict the market Value of a player**.
- Classification — **group players into performance tiers (Low, Mid, High, Elite)** .
- Model Validation — evaluate all models rigorously using Cross-Validation techniques

- **Each group MUST consist of FIVE students**
- **No cheating.** It will be heavily penalized for both parties.
- **Deadline Friday 1/5/2026 11:59 PM**

Dataset

Use the FIFA Players Dataset :

https://drive.google.com/drive/folders/13Qb935zhUWRxn4yxMRqbsixXnEGLkhbC?usp=drive_link

The dataset contains the following columns:

- **Name:** Full name of the player
- **Country:** Nationality of the player
- **Position:** Playing position on the field, e.g. ST, GK, CB
- **Age:** Age of the player in years
- **Overall_Rating:** FIFA overall rating
- **Future Potential:** Projected peak rating for the player
- **Team:** Current club or national team
- **Value Per M$:** Market value in millions USD
- **Total_Stats Score:** Composite score aggregating all in-game skill attributes

Task 1 : Exploratory Data Analysis (EDA)

Apply all EDA Python techniques you learned in the labs to answer the following questions:

- Are there any missing values? How many, and in which columns?
- What does the distribution of Value Per M$ look like? Is it approximately normal or skewed? Compute the skewness.
- Which numerical features seem most related to Value Per M$?
- What is the average Overall_Rating per Position?

Task 2 : Data Preprocessing

Apply the same preprocessing steps for all models. You must:

- **Split the dataset into Train and Test sets (80/20) before applying any preprocessing step.**
- **Encode categorical columns**
- **Scale numerical features**
- **Detect and handle outliers**

Task 3 : Create Classification Target (Logistic Regression + Naïve Bayes)

Use the dataset to classify players into 4 performance tiers **based on Overall_Rating or Total_Stats Score**

- Decide on the thresholds for each category.
- Justify your thresholds — use the data distribution (percentiles or histogram) to support your decision.
- Show how many players fall into each category after splitting.
- Plot a bar chart showing the class distribution. Is it balanced or imbalanced?

Task 4 : Model 1: Polynomial Regression

Train regression models to predict the player market value.

- Train a baseline Linear Regression model and evaluate it using: MAE, MSE, RMSE, and R^2 score on both train and test sets.
- Use PolynomialFeatures **to expand the numerical features only** for degree = 1, 2, 3, and 4 if needed.
- For each degree, record Train R^2 and Test R^2. Plot degree vs. performance. What trend do you observe?
- Which degree gives the best generalization? Justify using the train/test gap.

Regularization (Ridge and Lasso)

Apply regularization on the best polynomial degree to control overfitting.

- **Ridge (L2):** Sweep alpha over a log-spaced range. Report Train and Test RMSE for each.
- **Lasso (L1):** Do the same.
- Plot both curves (alpha on log x-axis vs. Test RMSE). Mark the best alpha for each.
- Compare Ridge vs. Lasso: which performs better on this dataset, and why?
- Did Lasso zero out any features? If yes, which ones and what does that imply?

Task 5 : Model 2: Logistic Regression

Train a Logistic Regression classifier to predict the player performance tier (Low / Mid / High / Elite). Overall_Rating must NOT be included as a feature.

- Train a baseline Logistic Regression model with default settings.
- Evaluate using: Accuracy, Precision, Recall, F1-score for each class.
- Show and explain the Confusion Matrix as a heatmap.

Regularization in Logistic Regression

In Logistic Regression, the parameter C controls regularization strength (smaller C = stronger penalty). Sweep C over a log-spaced range (e.g. 10^{-3} to 10^{3}):

- For each value of C, record Train Accuracy and Test Accuracy.
- Plot Train Accuracy and Test Accuracy vs. C (use log scale on x-axis). Identify the best C value.
- **Compare:** penalty='l1' (saga solver) vs. penalty='l2' (lbfgs solver) at the best C. Which regularization performs better?

Task 6 : Model 3: Naïve Bayes Classification

Train Naïve Bayes classifiers to predict the player performance tier.

Naïve Bayes Models

Implement the following Naïve Bayes variants and compare their performance:

- **GaussianNB** — train on numerical features only (Age, Future Potential, Total_Stats Score)
- **BernoulliNB** — train on the full OHE-encoded feature set (binary columns)
- **ComplementNB** — train on the full OHE-encoded feature set (shifted to non-negative)

For each model:

- Evaluate using: Accuracy, Precision, Recall, F1-score.
- Display the Confusion Matrix as a heatmap.
- Which Naïve Bayes variant is most appropriate for this dataset? Justify your answer based on the nature of the features.
- **Scaling Sensitivity: Train GaussianNB with and without StandardScaler. Does scaling affect its performance? Explain why or why not based on how Naïve Bayes works.**

Task 7 : Model Evaluation with Cross-Validation

Instead of relying on a single train/test split, use Cross-Validation to obtain more reliable performance estimates for your models.

A. K-Fold Cross-Validation (Regression)

- Apply 5-fold cross-validation to your best regression model from Task 4.
- Report the mean RMSE, standard deviation, and all individual fold scores.
- Plot a fold-by-fold RMSE bar chart with a mean line.

B. Stratified K-Fold Cross-Validation (Classification)

- Apply StratifiedKFold (5 splits) to your best Logistic Regression model and your best Naïve Bayes variant.
- Report mean accuracy and standard deviation for each.
- Plot a grouped bar chart comparing Logistic Regression vs. Naïve Bayes fold-by-fold accuracy.
- Which model performs better and which is more stable across folds?

Task 8 : Analysis and Discussion

(Answer the following questions inside your Notebook)

8.1 Model Comparison

- Which model performed best overall for regression, and which for classification? Justify briefly.
- Is classification easier or harder than regression on this dataset? Why?

8.2 Regularization Analysis

- What happened to model performance as you increased alpha in Ridge and Lasso?
- Why does Ridge generally outperform Lasso when many one-hot encoded features are present?

Good Luck 🤗